

LEAKMIND — PAC REVIEW PRESENTATION BRIEF

PHASES 1–11 COMPLETE
Academic Defense & Committee Review

An Explainable Multi-Evidence Framework for Data-Leakage Detection and Policy-Aware Response

Repository: github.com/akshad-n/Leakmind

SLIDE 1: PROJECT TITLE AND OBJECTIVES

Project Title: LeakMind: An Explainable Multi-Evidence Framework for Data-Leakage Detection and Policy-Aware Response

Primary Objective: Build an integrated security framework that detects, attributes, explains, and mitigates multi-stage enterprise data exfiltration by fusing heterogeneous evidence sources.

Key Specific Objectives:

- Ingest and preprocess multi-source enterprise logs (Logon, Device, File, HTTP, Email).
- Combine 4 heterogeneous evidence streams: Behavior AI, Sensitive Data AI, Provenance Graph, and Temporal Correlation.
- Eliminate false alarms generated by single-point anomaly detectors on benign employee work.
- Deliver local game-theoretic explainability using SHAP (Shapley Additive exPlanations) for SOC analysts.
- Enforce automated, policy-governed mitigations: ALLOW (<40%), MONITOR (40–70%), ALERT (70–90%), BLOCK ($\geq 90\%$).

SLIDE 2: PROBLEM STATEMENT AND PROPOSED SOLUTION

The Core Problem in Modern Enterprise DLP:

- **High False Positive Rates (FPR):** Conventional UEBA detectors flag routine employee behaviors (working late, bulk legitimate file copy) as insider attacks.
- **Multi-Stage Detection Blindspots:** Signature/regex DLP tools cannot catch low-and-slow exfiltration sequences where legitimate access is followed hours later by USB/cloud staging.
- **Black-Box Opacity:** Deep learning models lack auditable explanations needed for legal compliance and SOC analyst trust.

LeakMind's Proposed Solution:

- **Multi-Evidence Reasoning:** Concurrently evaluates user behavior, content sensitivity, provenance reachability, and temporal kill chains.
- **Supervised XGBoost Fusion:** Captures non-linear evidence interactions to output calibrated risk probabilities (0–100%).
- **Game-Theoretic SHAP Attributions:** Decomposes each decision into exact positive risk drivers and negative mitigating factors.
- **Configurable Policy Governance:** Translates continuous risk into 4 operational security responses.

SLIDE 3: WORK COMPLETED TILL DATE (PHASES 1–11)

- **Phases 1–3 (Data & Features):** Ingested CERT r4.2/r1 logs, engineered 18 behavioral/context features over daily sliding windows.
- **Phase 4 (Behavior AI):** Developed Context-Aware Isolation Forest with role clearance and after-hours frequency weighting.
- **Phase 5 (Sensitive Data AI):** Built dual-stage hybrid NLP classifier (TF-IDF + compiled regexes for keys, SSN, IBAN; 88% F1-score).
- **Phase 6 (Provenance Graph):** Implemented NetworkX directed multigraph tracking process-to-file-to-egress reachability.
- **Phase 7 (Temporal Correlator):** Built multi-stage kill chain automaton scoring sequential attack progression with time decay.
- **Phase 8 (Risk Fusion):** Trained and benchmarked XGBoost against Random Forest and Logistic Regression on 140 ground-truth scenarios.
- **Phase 9 (SHAP Explainability):** Implemented TreeSHAP generating waterfall attributions and natural-language narrative reports.
- **Phase 10 (Policy Engine):** Built 4-tier decision gate (ALLOW, MONITOR, ALERT, BLOCK) with deterministic credential override rules.
- **Phase 11 (Pipeline & Dashboard):** Built master orchestrator and an interactive 9-view Streamlit dashboard for academic demonstration.

SLIDE 4: METHODOLOGY AND SYSTEM ARCHITECTURE

Data Flow & Architecture Pipeline:

- **Stage 1 (Preprocessing):** Raw logs → Sessionization → 18 Context-Aware Features.
- **Stage 2 (Four Evidence Streams):** Behavior AI (0–100) + Sensitive Data AI (0–100) + Provenance Graph (0–100) + Temporal Correlator (0–100).
- **Stage 3 (Risk Fusion):** XGBoost evaluates 10-feature vector → Calibrated Risk Score (0.0% – 100.0%).
- **Stage 4 (SHAP Explainability):** TreeExplainer computes exact local attributions $\phi_i(x)$ for positive risk drivers and mitigators.
- **Stage 5 (Policy Enforcement):** Evaluates risk, sensitivity level, destination, and context → Triggers ALLOW, MONITOR, ALERT, or BLOCK.

SLIDE 5: IMPLEMENTATION AND EXPERIMENTAL RESULTS

Empirical Model Benchmarks (Holdout Test Cohort, N = 35):

- **Logistic Regression:** Accuracy: 1.0000 | Precision: 1.0000 | Recall: 1.0000 | ROC-AUC: 1.0000
- **Random Forest:** Accuracy: 1.0000 | Precision: 1.0000 | Recall: 1.0000 | ROC-AUC: 1.0000
- **XGBoost (Production):** Accuracy: 1.0000 | Precision: 1.0000 | Recall: 1.0000 | ROC-AUC: 1.0000 (native TreeSHAP support)

Incremental Evidence Ablation Findings:

- **Behavior AI Only:** 97.14% Accuracy, 94.44% Precision, **5.56% False Positive Rate (1 false alarm on benign after-hours worker).**
 - **Behavior + Sensitive Data:** 100% Accuracy, 100% Precision, **0.00% False Positive Rate (0 false alarms).**
 - **All 4 Evidence Sources:** 100% Accuracy, 100% Precision, **0.00% FPR** — complete elimination of false alarms.
- Global Feature Importance (SHAP Ranking):** 1. Sensitivity Risk (35.1%) • 2. Leakage Chain (32.4%) • 3. Graph Risk (17.5%) • 4. Historical Deviation (9.0%) • 5. USB Activity (2.2%).

SLIDE 6: PROGRESS AGAINST PLANNED MILESTONES

Milestone	Scope & Core Deliverables	Status
Milestone 1	Dataset Ingestion, Preprocessing & Daily Feature Extraction	100% Complete
Milestone 2	Behavior AI (Isolation Forest) & Sensitive Data AI standalone engines	100% Complete
Milestone 3	Provenance Knowledge Graph & Multi-Stage Temporal Kill Chain Correlator	100% Complete
Milestone 4	Multi-Evidence XGBoost Risk Fusion & Supervised Benchmark Comparisons	100% Complete
Milestone 5	TreeSHAP Local Explainability & 4-Tier Automated Policy Engine	100% Complete
Milestone 6	Interactive Streamlit Dashboard & Controlled End-to-End Simulation Demo	100% Complete

SLIDE 7: PAC COMMITTEE FEEDBACK AND ACTIONS TAKEN

- **PAC Feedback 1:** *"Single-evidence anomaly detection yields too many false alarms during legitimate off-hours employee work."*
Action Taken: Integrated contextual weights (role, destination) and fused content sensitivity and temporal correlation, reducing FPR from 5.56% to 0.00%.
- **PAC Feedback 2:** *"Explainability must not rely on post-hoc black-box heuristics or hardcoded rules."*
Action Taken: Integrated game-theoretic TreeSHAP computing exact additive feature attributions with natural language narrative generation for each individual incident.
- **PAC Feedback 3:** *"Provide automated operational mitigation response rather than just a passive risk score."*
Action Taken: Implemented a 4-tier configurable Policy Engine (ALLOW, MONITOR, ALERT, BLOCK) with deterministic overrides for critical credential egress.
- **PAC Feedback 4:** *"Ensure models are portable across development laptops without mandatory retraining."*
Action Taken: Persisted self-contained, zero-retraining model bundles in saved_models/ with fitted estimators, scalers, and JSON configs.

SLIDE 8: CURRENT CHALLENGES AND PROPOSED SOLUTIONS

- **Challenge 1: Telemetry Sparsity in Benchmark Logs:** Public datasets (e.g. CERT) lack kernel-level system-call provenance edges.
Proposed Solution: Graceful Degradation Architecture — NetworkX graph traversal operates on available audit events and gracefully interfaces with DARPA TC Neo4j when available.
- **Challenge 2: SHAP Computation Overhead in Real-Time Streaming:** Real-time explainability can introduce latency.
Proposed Solution: Leveraged optimized C++ TreeSHAP implementation on gradient-boosted decision trees, computing local attributions in under 15ms per session.
- **Challenge 3: Avoiding Rigid Hard-Coded Policy Thresholds:** Fixed thresholds fail across different corporate risk postures.
Proposed Solution: Externalized policy governance into JSON configuration with an interactive threshold adjustment slider in the Streamlit UI.

SLIDE 9: WORK PLANNED FOR THE NEXT PHASE

- **Task 1: Live Cloud Deployment:** Host the Streamlit dashboard on Streamlit Community Cloud for public live evaluation.
- **Task 2: Real-Time Stream Ingestion:** Connect pipeline with live streaming ingestion brokers (Kafka / Syslog / Windows Event Forwarding agent).
- **Task 3: Graph Neural Network (GNN) Exploration:** Conduct comparative ablation testing between GNN embeddings and shortest-path graph traversal.
- **Task 4: Active Learning & Feedback Loop:** Implement analyst feedback storage (analyst_feedback.json) to dynamically update SHAP feature priors.

SLIDE 10: EXPECTED FINAL OUTCOME AND DELIVERABLES

- **Deliverable 1: Core Python Framework:** Modular, tested, and containerized LeakMind detection pipeline.
- **Deliverable 2: Interactive Demonstration Dashboard:** 9-view academic Streamlit dashboard for real-time monitoring and simulated incident investigation.
- **Deliverable 3: Empirical Benchmark Reports:** Comprehensive evaluation reports with ablation tables, ROC curves, and SHAP waterfalls.
- **Deliverable 4: Research Publications & Documentation:** Research paper draft and complete technical documentation ready for academic publication/thesis defense.